

DEFIANCE, OH, USA

WMO: 725254

Lat: 41.338N

Lon: 84.429W

Elev: 707

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 04851

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	5.3	-7.6	3.8	0.8	-2.0	5.1	6.8	22.3	30.8	20.2	30.1	5.2	240	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.8	73.6	88.4	72.6	85.8	71.0	77.0	86.9	75.3	84.1	73.8	81.9	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	127.9	81.8	72.5	123.4	80.5	71.4	118.9	79.2	40.7	86.7	39.1	84.1	37.7	81.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.7	17.7	16.0	DB	-6.8	94.4	6.7	3.8	-11.7	97.2	-15.6	99.4	-19.4	101.6	-24.2	104.3	(4)
			WB	-7.4	79.1	6.5	2.0	-12.1	80.5	-15.9	81.6	-19.6	82.7	-24.3	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	25.6	28.2	37.9	50.2	61.2	70.2	73.7	71.6	65.3	53.6	41.8	31.1	(6)
		DBStd	19.17	11.89	11.01	10.87	9.17	8.52	6.56	5.72	4.98	7.43	8.95	9.40	10.62	(7)
		HDD50	2822	758	614	402	107	9	0	0	1	64	277	590		(8)
		HDD65	5956	1221	1031	840	450	177	26	3	85	370	696	1050		(9)
		CDD50	3188	1	3	28	114	357	605	736	670	461	175	32	6	(10)
		CDD65	847	0	0	1	7	61	182	274	212	95	15	0	0	(11)
		CDH74	7784	0	0	16	90	603	1628	2648	1766	892	138	3	0	(12)
		CDH80	2570	0	0	3	10	172	558	1006	513	281	27	0	0	(13)
(14)	Wind	WSAvg	6.8	8.0	7.8	8.0	8.6	7.0	5.9	5.0	4.7	5.1	6.5	7.5	7.5	(14)
(15)	Precipitation	PrecAvg	35.5	2.4	2.2	2.4	3.3	3.9	3.6	3.7	3.5	3.0	2.7	2.5	2.4	(15)
(16)		PrecMax	43.9	5.2	5.5	5.0	5.9	6.4	6.9	6.8	10.2	5.9	6.9	5.3	5.5	(16)
(17)		PrecMin	26.7	0.8	0.5	0.7	0.7	1.3	1.0	1.8	0.9	0.5	0.3	1.0	0.5	(17)
(18)		PrecStd	4.5	1.1	1.3	1.0	1.3	1.7	1.4	1.5	2.3	1.7	1.8	1.1	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	59.5	63.6	75.2	81.6	88.5	93.0	95.7	91.3	90.9	83.8	71.5	63.0		(19)
(20)		MCWB	55.2	53.7	63.1	64.9	70.1	73.3	76.2	73.8	72.4	69.8	59.6	58.2		(20)
(21)		DB	52.7	54.9	66.3	76.3	84.5	89.7	91.2	88.1	87.5	77.7	64.8	56.4		(21)
(22)		MCWB	49.1	50.9	57.1	61.0	67.7	72.1	75.0	73.4	71.2	64.2	56.5	52.7		(22)
(23)		DB	47.0	48.3	61.0	72.0	81.2	86.1	88.7	85.2	82.6	73.1	61.2	50.9		(23)
(24)		MCWB	43.5	44.1	53.4	58.9	66.5	70.7	73.5	71.8	68.7	62.8	54.1	47.2		(24)
(25)		DB	41.1	42.9	55.2	66.6	76.6	82.6	85.8	82.4	79.3	69.0	56.7	45.3		(25)
(26)		MCWB	37.9	38.8	48.8	55.7	64.4	69.3	71.7	70.5	66.8	60.3	50.8	42.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	55.8	57.0	63.7	67.9	73.2	77.0	79.1	78.2	75.9	71.3	62.2	58.7		(27)
(28)		MCDB	60.0	60.9	75.0	77.1	83.8	88.5	89.5	87.4	86.7	81.1	68.5	62.4		(28)
(29)		WB	50.0	50.7	58.7	63.8	70.5	74.9	77.6	76.2	73.5	67.9	58.3	53.2		(29)
(30)		MCDB	52.2	55.0	65.2	72.2	80.7	84.9	88.2	84.2	82.6	74.5	63.3	55.7		(30)
(31)		WB	43.8	44.4	54.7	60.4	68.6	73.1	75.8	74.4	71.4	64.5	55.1	47.2		(31)
(32)		MCDB	47.1	48.0	59.9	69.0	77.7	82.8	85.1	81.6	79.6	70.4	60.6	50.2		(32)
(33)		WB	38.0	38.9	49.5	57.3	66.4	71.4	74.1	72.7	69.3	61.4	51.2	42.2		(33)
(34)		MCDB	41.0	42.5	55.0	65.9	74.4	79.6	82.2	79.5	76.1	68.2	56.2	45.3		(34)
(35)	Mean Daily Temperature Range	MDBR	13.9	14.9	17.5	20.7	20.6	19.9	20.4	20.0	21.7	19.5	15.8	12.7		(35)
(36)		MCDBR	17.8	21.0	25.0	27.5	25.5	23.6	22.3	22.4	25.8	24.4	22.0	17.7		(36)
(37)		MCWBR	15.1	17.4	17.1	15.8	12.7	10.3	8.8	10.1	11.8	13.5	15.6	15.0		(37)
(38)		MCDBR	17.0	20.4	22.2	23.6	22.0	19.6	19.5	18.4	20.5	20.2	19.2	16.5		(38)
(39)		MCWBR	15.3	17.8	16.9	16.0	12.5	10.0	9.0	9.6	10.4	13.1	15.4	15.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.335	0.352	0.401	0.434	0.446	0.439	0.437	0.396	0.355	0.324	0.307		(40)
(41)		taud	2.382	2.284	2.335	2.214	2.170	2.181	2.219	2.210	2.303	2.405	2.443	2.412		(41)
(42)		Ebn at Noon	268	274	284	276	268	264	265	262	266	265	259	257		(42)
(43)		Edh at Noon	26	34	36	43	47	46	44	43	37	30	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	498	794	1116	1472	1709	1911	1926	1691	1383	919	590	417		(44)
(45)		RadStd	55	107	99	131	154	130	104	76	115	80	81	46		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page